Alex Carroll

MIT Center for Transportation and Logistics
Cambridge, MA
alex629@mit.edu | amcmit.github.io | linkedin.com/in/alex-carroll629

Research Agenda

My research examines how AI and operations research can improve supply-chain decisions under uncertainty across three linked areas. First, I develop agentic procurement-intelligence systems that combine hybrid retrieval, knowledge graphs, and bounded verification across heterogeneous enterprise data; this work includes a multinational supplier-intelligence deployment and a hyperscaler collaboration on agentic bill-of-materials knowledge graphs. Second, I apply reinforcement learning, dynamic programming, and ensemble learning to freight pricing, routing, market sizing, and decarbonization. Third, I map supply networks in emerging technologies, especially quantum computing, using self-hosted search, local language models, and provenance-aware property graphs to identify chokepoints and single-source dependencies.

Education

Massachusetts Institute of Technology, Cambridge, MA — 2026 – Present
PhD in Transportation

Massachusetts Institute of Technology, Cambridge, MA — 2025 – 2026
Master of Science in Transportation

Massachusetts Institute of Technology, Cambridge, MA — 2023 – 2024
Master of Applied Science in Supply Chain Management

University of Illinois at Urbana-Champaign, Champaign, IL — 2021 – 2023
Master of Business Administration

The University of Texas at Austin, Austin, TX — 2011 – 2016
Bachelor of Arts

Research Experience

Massachusetts Institute of Technology, Center for Transportation and Logistics — 2025 – Present
Research Assistant and Doctoral Researcher

- Developed and benchmarked five enterprise retrieval architectures across 80 curated queries, evaluating exact-match accuracy, F1, multi-source recall, abstention accuracy, latency, and cache performance.
- Built supplier-intelligence and knowledge-graph workflows spanning RFx documents, contracts, spend data, and supplier records across vector, relational, and graph stores, with auditable provenance and confidence-gated verification.
- Formulated constrained Markov decision processes and stochastic simulators for freight pricing and routing, implementing and evaluating value-based and actor-critic methods including PPO, A2C, SAC, DQN, and TD3 under capacity, service-time, and cost constraints.

- Built ensemble market-sizing and forecasting pipelines across 12,587 freight lanes using XGBoost, CatBoost, geospatial features, and SHAP analysis; quantified service feasibility under 2030, 2035, and 2040 operating scenarios.

Publications

Refereed Journal Articles

- **Carroll, A.**, "Fast Barge, Smart Prices: Deep Reinforcement Learning for Mississippi Container Service," *Annals of Operations Research*, Special Issue on Innovations in Maritime and Port Logistics: Optimization and Data-Driven Decision Making, 2026.
- **Carroll, A.**, and Dugundji, E., "A Multi-Filter Market-Sizing Framework for Mississippi River Containerized Freight: Assessing Serviceable Obtainable Demand through Ensemble Machine Learning," *Transportation Research Record*. Submitted March 2026; in revision.

Journal Articles Under Review

- **Carroll, A.**, and Dugundji, E., "Agentic Generative AI for Procurement Intelligence: A Source-Routing Evaluation," *International Journal of Production Research*, Special Issue on The Agentic Supply Chain: Entering a New Era in AI in Supply Chain Management. Submitted June 2026; under review.

Refereed Conference Papers and Proceedings

- **Carroll, A.**, and Dugundji, E., "Performance Engineering of Retrieval Architectures for Enterprise Supply-Chain Question Answering," *IEEE High Performance Extreme Computing Conference* (HPEC), September 2026.
- Power, K., and **Carroll, A.**, "Enhancing Sentiment Classification of E-commerce Reviews for Actionable Insights Using Large Language Models," *IEEE High Performance Extreme Computing Conference* (Virtual), September 2025.
- **Carroll, A.**, Egar, T., Koch, T., and Dugundji, E., "Drop Trailer Forecasting in Volatile Networks," *Procedia Computer Science*, Elsevier. Proceedings of the 15th International Conference on Ambient Systems, Networks and Technologies (ANT), April 2024.

Other Publications

- **Carroll, A.**, Egar, T., Koch, T., and Dugundji, E., "Drop Trailer Management in Volatile Networks," *Supply Chain Management Review*, June 25, 2025.

Talks and Presentations

- "Performance Engineering of Retrieval Architectures for Enterprise Supply-Chain Question Answering," IEEE High Performance Extreme Computing Conference (HPEC), September 2026.
- "Optimizing Drop Trailer Networks: Advanced Analytics for Efficient Supply Chains," Session B1-TS05A, SIG B1: Supply Chain Management: Supply Chain and Logistics Management, World Conference on Transport Research (WCTR 2026), Toulouse, France, July 2026. *Session chair and presenter.* Invited to submit an extended manuscript to *Transportation Research Record* following conference peer review.

- "Strategic Investments in Freight Decarbonization: Quantifying the Benefits of Zero-Emission Corridor Deployment and Strategic Rail Modal Shifts," Session B3-TS08A, SIG B3: Freight Transport Operations and Intermodality: Transforming Freight Transport for a Sustainable Future, World Conference on Transport Research (WCTR 2026), Toulouse, France, July 2026.
- "Deep Reinforcement Learning for Mississippi Container Service," Poster Session, INFORMS Transportation Science and Logistics Society Conference, MIT, Cambridge, MA, July 2026.
- "The Price is Right: Deep Reinforcement Learning for Dynamic Pricing in Freight Markets," The AI-Driven Supply Chain: Advanced Training for Next-Gen Leaders, MIT, Cambridge, MA, July 2026. *Invited instructor.*
- "Review Mining with Transformers and LLMs," The AI-Driven Supply Chain: Advanced Training for Next-Gen Leaders, MIT, Cambridge, MA, July 2026. *Invited instructor.*
- "AI-Powered Automation in Ports," Workshop on AI-Powered Automation in Ports, Rutgers University DIMACS Center, Piscataway, NJ, March 2026.
- "Assessing the Total Addressable Market of Barge Service on the Mississippi River," Transportation Research Board Annual Meeting, Washington, DC, January 2026.
- "From Silos to Agentic Systems: Leveraging Agentic AI for Supplier Intelligence," MIT Center for Transportation and Logistics Agentic AI Roundtable, December 2025.
- "Advanced Analytics for Efficient Supply Chains," 7th Interdisciplinary Conference on Production, Logistics and Traffic, Darmstadt, Germany, March 2025.
- "Drop Trailer Management in Volatile Networks," SCALE Expo, MIT, January 2024.

Teaching Experience

- The AI-Driven Supply Chain: Advanced Training for Next-Gen Leaders, MIT, July 2026, Instructor.
- SCM.256 Data Science and Machine Learning for Supply Chain Management, Spring 2026, Teaching Assistant.
- Research Mentor, GCLOG Capstone, MIT CTL, July 2025 – December 2025. Provided technical and methodological guidance on problem formulation, pricing models, and empirical analysis for the student research paper "Monetizing Modal Innovation: Pricing Next-Generation Barge Services."
- AI in Supply Chain and Logistics Management, MIT CTL Summer School, July 2025, Teaching Assistant.
- SCM.256 Data Science and Machine Learning for Supply Chain Management, Spring 2025, Industry Partner Project Sponsor.

Professional Experience

Amazon, Tokyo, Japan — Jun 2024 – Aug 2025
Senior Operations Manager, Amazon Logistics

- Increased throughput per labor hour by 11% quarter over quarter, led daily processing of more than 110,000 packages across 300+ associates, and achieved the region's highest-ranked site performance in Q3 and Q4.
- Directed implementation of an AI-driven sortation process, reducing mis-sorts by 9% and saving a projected 11,500 labor hours annually across multiple cycle shifts.

- Streamlined inbound, sortation, and pick workflows, improving work-in-process compliance by 10% and contributing to a 5% improvement in on-time outbound delivery performance.
- Launched safety and productivity initiatives that reduced incident rates by 15% and increased employee engagement scores by 22% from Q2 to Q4 2024.
- Collaborated with academic partners and internal research teams on data-driven process improvement and applied machine learning initiatives.

Flock Freight, San Diego, CA — Jan 2022 – Aug 2023
Senior Analyst, Operations & Analytics

- Managed a dynamic pricing model providing real-time quotes for more than 60% of revenue, optimizing cost, margin, conversion, and probability-based pooling estimates.
- Improved costing-model accuracy by 10% MAPE, representing a $6MM year-over-year improvement in cost savings and a $4.2MM increase in gross profit.
- Automated pricing processes, increasing productivity by 40% and shifting analyst capacity toward higher-value pricing strategy.
- Collaborated with engineering, data science, and product teams to implement the APE machine-learning pricing engine, reducing pricing labor hours by 40%.
- Built batch-rating and quoting automation for enterprise customers, removing workload equivalent to four quality-assurance associates.
- Directed daily pricing audits with Customer Success and Sales, using conversion, gross-profit, and cost-savings data to determine customer-specific strategies.

Zillow, Seattle, WA — Jun 2021 – Dec 2021
Pricing Analyst, Acquisitions

- Led valuation of more than $200MM in residential real estate using internal and external data, financial models, and risk analysis.
- Led Austin and San Antonio market research using Tableau, Excel, SQL, Python, and proprietary machine-learning tools.
- Produced weekly market-intelligence analyses and improved team valuation accuracy by 2.7% MAPE through sales-funnel outlier analysis.
- Developed five adopted proposals across Sales, Data Science, and Market specialization teams, reducing case time per analyst by 18%.

Arrive Logistics, Austin, TX — Feb 2017 – Jun 2021
Pricing Analyst, Strategy

- Developed strategic and tactical pricing programs that helped increase contract revenue by 320% from January 2018 to January 2021.
- Led concurrent deals ranging from $1MM to $30MM and closed more than $150MM in revenue during tenure.
- Analyzed industry trends, competitors, and market dynamics to provide pricing and strategy recommendations to enterprise clients and internal stakeholders.

Technical Skills

- **Optimization and Decision Science:** Deep Reinforcement Learning (PPO, A2C, SAC, DQN, TD3; adversarial and robust RL), Dynamic Programming, Markov Decision Processes and Constrained MDPs, Mixed-Integer Linear Programming, Vehicle Routing and Network Flow, Dynamic Pricing and Revenue Management, Stochastic Simulation
- **Statistical and Predictive Modeling:** Gradient-Boosted Trees (XGBoost, CatBoost), Ensemble Machine Learning, Time-Series Forecasting (ARIMA/SARIMA, exponential smoothing, Prophet), Demand Modeling, Synthetic Data Generation, Model Explainability (SHAP), Traffic Microsimulation (SUMO), Geospatial and Network Analysis
- **AI Systems and NLP:** Agentic and Multi-Agent Systems, Retrieval-Augmented Generation, GraphRAG and Hybrid Retrieval, Multi-Source Routing, Abstention and Verifier Gating, Retrieval-System Evaluation, Knowledge-Graph Construction, Entity Resolution, Enterprise Document and OSINT Information Extraction, Transformer Fine-Tuning, Embedding and Vector Search, Local LLM Deployment
- **Programming and Frameworks:** Python (Pandas, NumPy, SciPy, Scikit-learn, Statsmodels, XGBoost, CatBoost, PyTorch, TensorFlow/Keras), SQL, Cypher, Bash, Git, LaTeX; Gurobi, Stable-Baselines3, Gymnasium, NetworkX, LangGraph, Hugging Face Transformers
- **Data and Infrastructure:** Neo4j, Amazon Neptune, FAISS, Pinecone, ChromaDB, SQLite, Docker, AWS (S3, EC2, SageMaker), OpenAI and Anthropic APIs, Ollama and local LLM serving
- **Evaluation, Visualization, and Reporting:** Experimental Benchmarking, Exact-Match and F1 Evaluation, Retrieval Recall, Latency and Cache Performance Profiling, Matplotlib, Seaborn, Power BI, Tableau, Excel

Languages

English (Native) | Japanese (Intermediate) | Spanish (Basic)

Service and Community

Session Chair, SIG B1: Supply Chain Management, World Conference on Transport Research (WCTR 2026)
MIT Science Policy Initiative, Executive Visit Days Participant
Austin-Oita Sister City Committee Member
Japan Society of Boston Member
Keep Austin Beautiful Member

Professional Memberships

INFORMS
World Conference on Transport Research Society